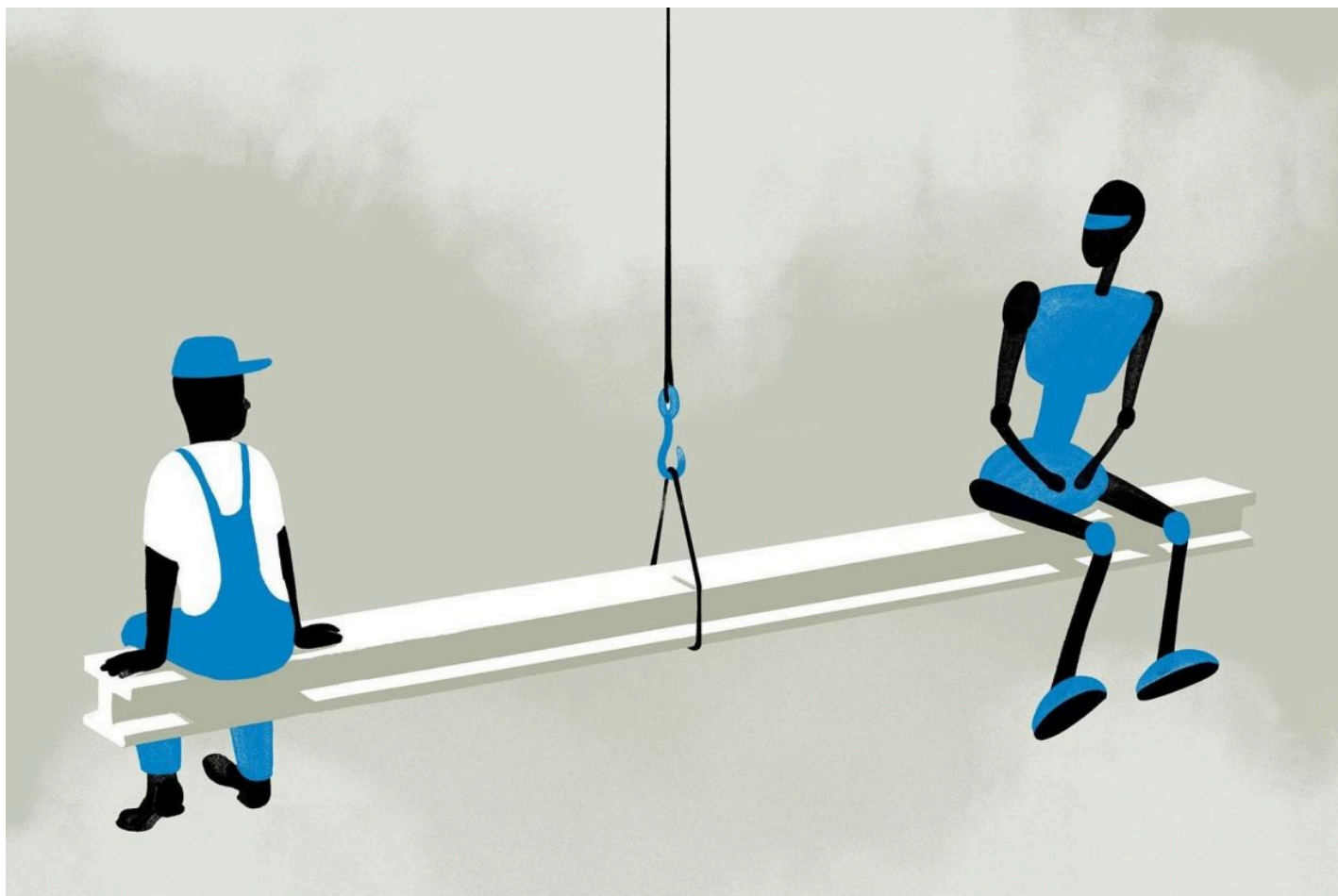


UW ECONOMICS SOCIETY



[HOME](#) [ABOUT US](#) [EXECUTIVE TEAM](#) [ORIGINAL CONTENT](#) [FEATURES](#) [STAY CONNECTED](#)



ORIGINAL ARTICLES September 24, 2020 • 2 Comments



Zuleykha
Gasimova

Could COVID-19 Bring Further Incentives for Automation?

Written By: Zuleykha Gasimova

The scale of impact COVID-19 has had on global economies calls to mind the effects of previous major recessions that have reshaped the economy and the views on governments' role in navigating crises.

As recessions are a part of the economic cycle, these events can foster overall economic efficiency by reallocating resources from less successful first to more successful and productive ones, creating an opportunity to innovate and become more efficient. However, the periods of economic downturn bring increased unemployment and permanent job losses as a result of firm incentives to cut labour costs and automate.

The data about labour market changes demonstrates that while non-routine employment has been relatively stable during the economic downturns, all of the permanent routine job losses in Canada have occurred during the last three recessions. This evidence asks for examining how recessions play a critical role in job polarization and reallocation in Canada more closely.

What is the link between recessions and automation?

One explanation to consider is that firms have higher incentives to engage in productivity-increasing activities such as technological upgrading, employee training and reorganizing operations during recessions due to the fact that the opportunity costs of such activities that slow-down production are lower.

Multiple research findings have demonstrated that during the recovery from the Great Recession the emphasis on hiring higher-skilled workers was mostly driven by technological changes. The data patterns suggest changes to labour demand for routine jobs disproportionately occurred in the harder-hit states and this was also correlated with major capital investments in automation technology. Hershbein and Kahn (2018)

Examining Recessions and Automation in Canada

In his paper, Professor Joel Blit has looked at three recession periods (1992 recession, 2009 recession and 2016 economic downturn) and the changes in routine job employment. Figure 1 below demonstrates the

monthly employment per capita for routine and non-routine jobs, where the examined recession periods are shaded in grey.

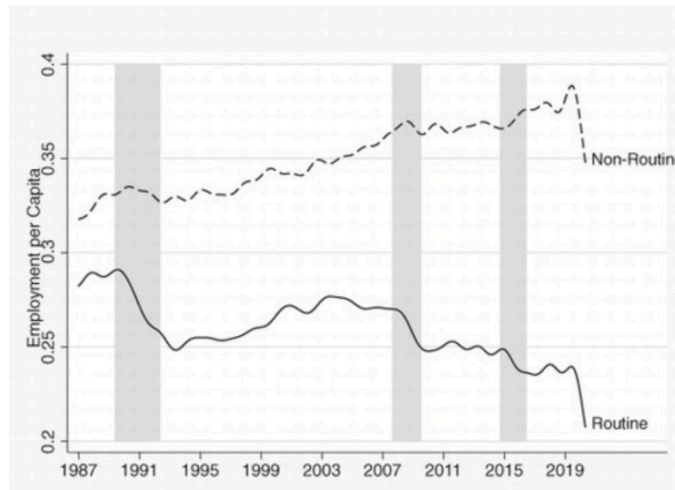


Figure 1: Monthly Employment in Routine and Non-Routine jobs between January 1987 and May 2020

Note: Blit, J (2020)

It is apparent from the data that there is a fall in routine employment during the economic downturn periods. For the period between January 1987 and December 2019, routine employment per capita fell by 5.4 percentage points while non-routine employment per capita increased by 5.6 percentage points.

When looking at each of the three recessions periods more closely, there is a pattern that routine jobs account for most of the job losses during recessions while non-routine jobs remain stable.

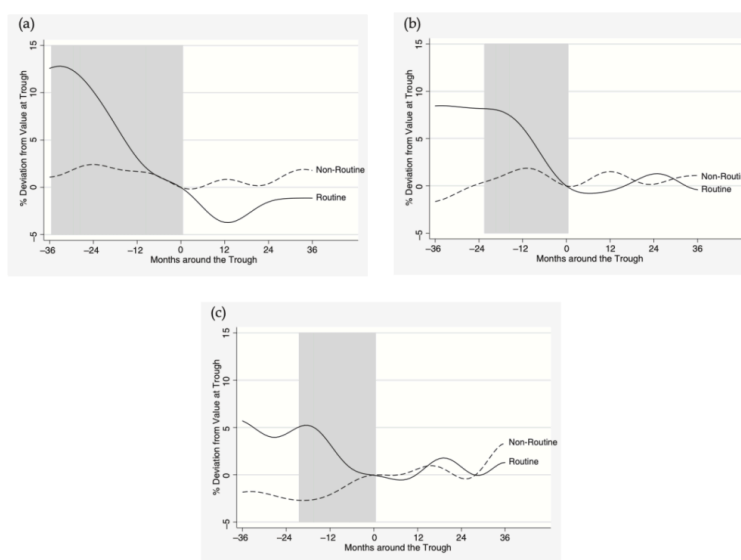


Figure 2: Monthly Employment in Routine and Non-Routine jobs for the (a) 1992, (b) 2009, and (c) 2016 Recessions

Note: Blit, J (2020)

These findings suggest that economic recessions are even more impactful in promoting automation and productivity in the Canadian economy in comparison with the United States.

How is the COVID Recession different?

The self-imposed measures to limit the economic activity in response to the rapid spread of the virus have led to a recession with longer-lasting effects on both demand and supply sides. A fall in wages and earnings along with the anxiety and uncertainty about potential new waves of COVID-19 is driving consumer and business spending down. On the supply side, many businesses and organizations might see this as an opportunity to automate and reorganize their operations motivated by not only economic considerations but health incentives as well. Given the unique conditions to protect the health of the workers by limiting contact, we have already seen that most jobs adapted a remote, work-from-home schedule even after some reopenings took place in May. In this regard information and communication technologies, AI, have played a critical role in enabling businesses to safely operate while complying with social distancing procedures.

Yet still, there are certain types of jobs that cannot be performed remotely and there will be greater COVID-related health incentives to implement automation technologies for these occupations.

To examine industry-specific incentives to automate, Professor Blit has studied the level of worker exposure measured by proximity and face-to-face interaction variables for each occupation in the given industries. The feasibility variable is obtained by categorizing occupations into non-routine and routine groups to measure how feasible it would be for a specific industry to automate.

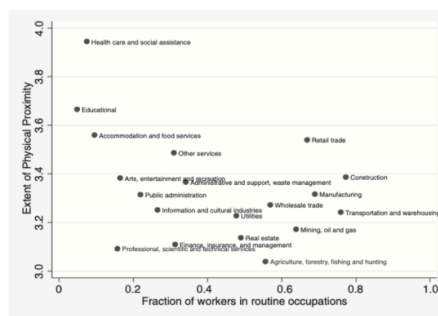


Figure 3: Extent of Physical Proximity and Fraction of Tasks that Are Routine for 19 Different NAICS Industries

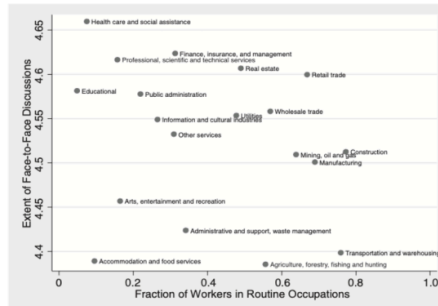


Figure 4: Extent of Face-to-Face Discussions and Fraction of Tasks that Are Routine for 19 Different NAICS Industries

Note: Blit, J (2020)

As can be seen from Figures 3 and 4, the industries on the upper right side such as retail trade, construction, wholesale and manufacturing are most likely to experience automation due to the COVID pandemic. In addition to these industries requiring many routine tasks that could be feasible to automate, there are higher health risks because of COVID exposure.

What are some policy implications to consider?

When it comes to the government response to the economic downturn induced by the COVID crisis we have seen policies to support businesses and organizations, wage subsidies and financial support packages. When explaining how the Canada Emergency Wage Subsidy might reduce the firm incentives to automate and reorganize, professor Blit notes that this policy is reducing the cost of labour while the opportunity costs to reorganize firm behaviour is lower. To see this, think about how profitable it would be to automate tasks that were being performed by a worker earning \$20 per hour versus when there is a 75% wage subsidy in place. The possibility to automate and reorganize efficiently in certain industries could be missed if the opportunity costs to upgrade or the competition with other firms is too high again by the time the government subsidy has ended.

While automation and implementation of innovative information and communication technologies during COVID-19 is an opportunity to foster overall economic growth, it can also result in greater social and economic inequality. Many less-educated, lower-income workers could either face higher health risks because of the lack of opportunities to work from home or socially-distance at work, or their jobs could get replaced as a result of the automation and reorganization of business operations.

Joel Blit advocates for a balanced policy approach moving forward when it comes to supporting businesses and industries to survive the crisis and to innovate and grow. While the industries as health care, education, accommodation and food services might need financial support the most, allowing other industries to reorganize and reallocate from less to more efficient firms would strengthen the overall productivity-growth that recessions can bring.

References:

Blit, J (2020), "Automation and Reallocation: Will COVID-19 Usher in the Future of Work?". Canadian Public Policy 46(S2): S192-S202.

<https://doi.org/10.3138/cpp.2020-065>

Bloom, D. and Prettnner, K (2020), The macroeconomic effects of automation and the role of COVID-19 in reinforcing their dynamics. VoxEU.

Retrieved from

<https://voxeu.org/article/covid-19-and-macroeconomic-effects-automation>

Hershbein, B., and L.B. Kahn (2018). "Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings." American Economic Review 108(7): 1737–72.
<https://doi.org/10.1257/aer.20161570>.

◆ Banner

Previous Post

Rising tuition: An economic crisis for universities and students alike

Next Post

Economics Newsletter – Week of September 21st

2 Comments on "Could COVID-19 Bring Further Incentives for Automation?"

ECONOMICS NEWSLETTER – WEEK OF SEPTEMBER 21ST | UW ECONOMICS SOCIETY SEPTEMBER 26, 2020 AT 10:42 AM

REPLY ↩

[...] Could COVID-19 Bring Further Incentives for Automation? By Zuleykha Gasimova [...]

DÉCOUVREZ DES OFFRES EXCEPTIONNELLES SUR NOTRE PHARMACIE EN LIGNE EN CLIQUANT ICI
APRIL 1, 2024 AT 8:28 AM



Acheter acticin à prix avantageux Belgique acticin : informations

REPLY ↩

essentielles pour les utilisateurs

acticin indication en vente au Maroc

commander acticin en ligne Belgique où trouver acticin en France

Livraison rapide de acticin à votre porte

acticin de marque disponible à l'achat en ligne

acheter acticin Belgique sans ordonnance médicale acheter imite en Belgique

Pharmacie en ligne française pour acheter du

imité en toute tranquillité

Acheter imite en ligne au Maroc en toute sécurité acticin abordable et de qualité

Obtenez votre imite sans attendre

Acheter de la imite en toute confidentialité

Sélection de acticin à prix abordable en Belgique

Acheter acticin avec livraison discrète France

Pharmacie en ligne pour acheter de la acticin en Belgique

Options d'achat de imite en Italie acticin : posologie recommandée et durée du traitement

Achat de imite pour les traitements de longue durée

acheter imite à Genève peut-on avoir acticin sans prescription médicale

imité en vente libre au Luxembourg Où acheter légalement de la imite en France

Trouver des offres spéciales sur la imite en ligne

Commander acticin sur internet France acticin et grossesse :

ce qu'il faut savoir

imité disponible sans prescription en ligne imite : guide d'achat en ligne pour les patients

acticin en ligne : avantages et inconvénients

imité : guide complet d'achat en ligne

imité sans prescription : comment choisir en toute sécurité ?

Obtenez des informations sur les interactions médicamenteuses de la imite en ligne

prix du imite Acheter acticin en ligne : guide étape par étape

pour les débutants

Acheter du acticin en ligne en toute tranquillité d'esprit acticin à prix compétitif en France

Leave a Reply

Your email address will not be published. Required fields are marked *

COMMENT

NAME *

EMAIL *

WEBSITE

POST COMMENT

☐ NOTIFY ME OF FOLLOW-UP COMMENTS BY EMAIL.

☐ NOTIFY ME OF NEW POSTS BY EMAIL.

